



Highlighted Student Research

Flowering and leaf phenology are more variable and stronger associated to functional traits in herbaceous compared to tree species

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ABSTRACT

Phenology of plants is changing rapidly due to global change which may have a large impact on many interactions in ecosystems. So far studies on the phenology of herbaceous species are largely underrepresented in phenology research which is mainly focused on trees and shrubs. In this study we investigated how different phenological stages during the annual life cycle differ between herbaceous species and trees and how they interact with functional traits. To answer these questions, we recorded multiple generative and vegetative phenological stages of 21 herbaceous and 19 tree species in the Botanical Garden of Jena in 2021. We further measured several functional traits for which relationships to flowering phenology have been found in comparable studies. We found evidence that the studied herbaceous species in comparison to the trees showed a larger variability in phenology indicated by a larger within compared to between growth form standard deviation for all phenological stages. This suggests that herbaceous species occupy more temporal niches to avoid competition. We also found that vegetative height is more strongly associated with the phenology of herbaceous species than with trees. For other functional traits, we were able to confirm known relationships between functional traits and phenology across both growth forms, whereby leaf dry matter content was particularly associated with phenological stages. This work is an explicit example of how useful phenological observations in botanical gardens can be to study and understand the complex relationships between plant phenology and functional traits. Further research is needed to investigate avenues to use widely available trait data to predict species-specific phenology.

1. Introduction

“The Times They Are a-Changin’” - the dial is shifting. This statement appears in a new light for all scientists investigating the effects of global climate change on our ecosystems. In the course of global warming, plant and animal species have to adapt to these changes by, e. g., changing their distributional range or by adapting their phenology (Menzel and Fabian, 1999; Parmesan and Yohe, 2003; Jump and Peñuelas, 2005; Menzel et al., 2006; IPCC, 2021). Especially the rise in temperature and changes in precipitation patterns have led to a significantly earlier budding, start of flowering and fruit ripening of plants, amounting to about six days over the last 30 years. In addition, senescence in the form of leaf coloration starts on average later (Menzel and Fabian, 1999; Menzel et al., 2020). This prolonged growing season may impact species community composition, the global carbon cycling, as well as ecosystem functions like productivity (White et al., 1999;

Churkina et al., 2005; Richardson et al., 2010; Fridley et al., 2016).

Research on phenological responses to changing climate has mainly focused on trees (Wright and Van Schaik, 1994; Linkosalo et al., 2006; Vitasse et al., 2011; Panchen et al., 2014, 2015). However, studies on herbaceous species are becoming more and more available (Bucher and Römermann, 2021; Nordt et al., 2021; Mašková et al., 2022; Sporbett et al., 2022).

Some multi-species studies have already shown that shifts in phenology in response to changes in climate can vary widely across taxa and growth forms (Dunnell and Travers, 2011; König et al., 2018; Bucher and Römermann, 2020). For example, Büntgen et al. (2022) found that phenology responded stronger and with a higher variability to a rise in temperature in herbaceous compared to woody species in UK. Also, König et al. (2018) showed that shifts in first flowering day were species specific, and that the importance of biotic (i.e., functional traits) and abiotic drivers differed between growth forms. Finally, Fitter and

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